

PAC-Bayesian Generalization Bounds for Weight-Decomposed Low-Rank Adaptation

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Abstract

Parameter-efficient fine-tuning (PEFT) methods rising in popularity as models increase in size. One paradigm of PEFT is weight decomposition, in which methods decompose neural network weight updates into magnitude and directional components. DoRA [1] and MAP [2] are representative methods of the weight decomposition paradigm, yet their generalization behavior remains poorly understood. We use PAC-Bayes to derive generalization bounds that explicitly capture the geometric structure of direction and magnitude updates for weight decomposition PEFT.

Our analysis introduces non-asymptotic bounds for von Mises-Fisher (vMF) distributions valid beyond the high-concentration regime and yields closed-form Kullback-Leibler (KL) complexity terms involving angular deviations on hyperspheres. We derive a data-dependent Gibbs posterior that induces a principled training objective, revealing how PAC-Bayes theory leads to cosine-based regularization.

We further establish a comparison theorem showing that column-wise methods such as DoRA distribute directional complexity across local angles, while global methods such as MAP concentrate it into a single rotation. From the comparison, we obtain a sharp characterization: DoRA achieves strictly tighter generalization bounds under sparse, localized updates, whereas MAP is favored under globally aligned transformations.

We formalize four testable predictions from the theory and describe a concrete experimental protocol — fine-tuning RoBERTa-base across six GLUE tasks under DoRA, MAP, and LoRA — designed to validate them in future work. Key predicted signatures include a positive Spearman rank correlation between the directional complexity estimator $\hat{C}_{\text{dir}}$ and test error, a sparse/dense regime crossover near $s/d \approx 0.5$, and substantially higher angular variance in MLP layers than attention layers, consistent with the layer-specific guidance derived from our framework.

1 Introduction

Large pretrained models are often adapted to downstream tasks using parameter-efficient fine-tuning (PEFT) methods, which modify only a small subset of parameters while preserving most of the pretrained weights. Among these approaches, recent methods such as DoRA [1] and MAP [2] adopt a *weight-decomposition* perspective, representing parameters in terms of their magnitudes and directions. Despite the empirical success of these methods, their generalization properties remain poorly understood.

A key challenge lies in understanding how different parameterizations affect the complexity of adaptation. DoRA performs *column-wise* updates, allowing each column of a weight matrix to rotate independently. MAP applies a *global* transformation that treats the entire flattened weight matrix as a single vector and rotates it as a whole. While these approaches differ geometrically, there is currently no theoretical framework that explains when or why one should generalize better than the other.

We address this gap through a PAC-Bayesian analysis of weight-decomposed adaptation. Our approach models directional updates using von Mises-Fisher (vMF) distributions on the hypersphere and derives generalization bounds that depend explicitly on angular deviations between pretrained and fine-tuned parameters. This leads to a notion of *directional complexity*, measured by cosine-based penalties that arise directly from KL divergence terms.

Our analysis yields several key insights. First, we derive non-asymptotic KL bounds for vMF distributions (theorems 3.6 and 3.7), enabling rigorous treatment of directional uncertainty without relying on high-concentration approximations. Second, we introduce a data-dependent Gibbs posterior (section 4) that yields a principled training objective, connecting PAC-Bayes theory to commonly used regularization schemes. Third, we establish a comparison theorem (theorem 3.10) showing that column-wise (local) methods distribute angular changes across many coordinates, while global methods compress them into a single rotation.

Our results lead to a sharp characterization of when each method is preferable. We show that DoRA achieves strictly tighter generalization bounds in regimes where updates are sparse and localized (theorems 3.12 and 3.14), whereas MAP is favored when updates are globally aligned. Through our work, we provide the first theoretical explanation for empirical phenomena observed across PEFT methods.

Finally, we empirically validate our theory by measuring PAC-Bayes complexity in practical fine-tuning settings (section 5). We show that angular complexity strongly correlates with generalization performance and that the predicted regimes—sparse versus dense updates—align with observed performance differences.

2 Background and Related Work

2.1 Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) [3] parameterizes the weight update $\Delta W = BA$ with low-rank matrices $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times d}$, $r \ll d$. DoRA [1] extends this by decomposing each weight column $w_j \in \mathbb{R}^d$ as $w_j = m_j u_j$ where $m_j \in \mathbb{R}_+$ is a learnable scalar magnitude and $u_j = w_j / \|w_j\|_2 \in \mathbb{S}^{d-1}$ is the unit-norm direction. The direction is updated via LoRA while the magnitude is trained directly. MAP [2] takes a global perspective by flattening the weight matrix $W \in \mathbb{R}^{d \times d}$ to $w \in \mathbb{R}^{d^2}$ and writing $w' = \alpha \hat{w} + \beta \Delta \hat{w}$ where $\hat{w}, \Delta \hat{w}$ are unit-norm vectors and $\alpha, \beta \in \mathbb{R}$ are learnable scalars.

2.2 PAC-Bayesian Generalization Theory

PAC-Bayes bounds [4, 5] provide high-probability guarantees on the generalization error of randomized (Gibbs) predictors. The central quantity is the KL divergence $\text{KL}(Q \| P)$ between a data-dependent posterior Q and a data-free prior P . Several prior works apply PAC-Bayes to fine-tuning [6, 7], but none have analyzed weight-decomposed methods through the lens of hyperspherical geometry.

2.3 Distributions on Hyperspheres

The von Mises-Fisher distribution $\text{vMF}(\mu, \kappa)$ on $\mathbb{S}^{d-1}$ has density

$$p(x \mid \mu, \kappa) = C_d(\kappa) \exp(\kappa \mu^\top x), \quad C_d(\kappa) = \frac{\kappa^{d/2-1}}{(2\pi)^{d/2} I_{d/2-1}(\kappa)},$$

where I_ν denotes the modified Bessel function of the first kind. As $\kappa \rightarrow \infty$ the distribution concentrates around μ ; at $\kappa = 0$ it reduces to the uniform distribution. The mean resultant length is $A_d(\kappa) = I_{d/2}(\kappa)/I_{d/2-1}(\kappa)$, which satisfies $A_d(\kappa) \in [0, 1)$ and is strictly increasing in κ .

3 PAC-Bayesian Analysis of Weight-Decomposed Adaptation

3.1 Setup and McAllester's Bound

Let $\mathcal{H}$ be a hypothesis space and $\mathcal{D}$ a data distribution over $\mathcal{Z}$. Let P be a prior over $\mathcal{H}$ chosen independently of the data, and Q a posterior learned from a sample $S = \{z_1, \dots, z_n\} \sim \mathcal{D}^n$. Assume the loss $\ell : \mathcal{H} \times \mathcal{Z} \rightarrow [0, 1]$. Define the population and empirical risks:

$$R(Q) := \mathbb{E}_{h \sim Q} \mathbb{E}_{z \sim \mathcal{D}} [\ell(h, z)], \quad \hat{R}(Q) := \mathbb{E}_{h \sim Q} \frac{1}{n} \sum_{i=1}^n \ell(h, z_i).$$

Theorem 3.1 (McAllester [4]). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the draw of S , for all posteriors Q :*

$$R(Q) \leq \hat{R}(Q) + \sqrt{\frac{\text{KL}(Q \| P) + \ln \frac{2\sqrt{n}}{\delta}}{2n}}.$$

The problem of bounding $R(Q)$ reduces to bounding $\text{KL}(Q \| P)$ under the structured parameterizations of DoRA and MAP.

3.2 Non-Asymptotic KL Bounds for vMF Distributions

We now avoid the high-concentration assumption and work with exact, non-asymptotic bounds. Lemmas 3.2 and 3.3 are well known.

Lemma 3.2 (vMF KL Divergence with Equal Concentration). *Let $Q_1 = \text{vMF}(\mu_1, \kappa)$ and $Q_0 = \text{vMF}(\mu_0, \kappa)$ be distributions on $\mathbb{S}^{d-1}$ with the same concentration $\kappa > 0$. Then:*

$$\text{KL}(Q_1 \| Q_0) = \kappa A_d(\kappa) (1 - \mu_1^\top \mu_0),$$

where $A_d(\kappa) = I_{d/2}(\kappa)/I_{d/2-1}(\kappa)$.

Lemma 3.3 (Uniform Bounds on $A_d(\kappa)$). *For all $\kappa > 0$ and $d \geq 2$:*

$$\frac{\kappa}{d + \kappa} \leq A_d(\kappa) \leq 1.$$

Theorem 3.4 (Non-Asymptotic Directional KL Bound). *Under the column-wise decomposition of DoRA with per-column vMF priors of concentration κ :*

$$\frac{\kappa^2}{d + \kappa} \sum_{j=1}^d (1 - \cos \theta_j) \leq C_{\text{dir}}^{\text{DoRA}} \leq \kappa \sum_{j=1}^d (1 - \cos \theta_j).$$

Under the global decomposition of MAP:

$$\frac{\kappa^2}{k + \kappa} (1 - \cos \Theta_{\text{global}}) \leq C_{\text{dir}}^{\text{MAP}} \leq \kappa (1 - \cos \Theta_{\text{global}}).$$

Proof. Apply theorem 3.2 to get $C_{\text{dir}}^{\text{DoRA}} = \sum_{j=1}^d \text{KL}(Q_{\text{dir},j} \| P_{\text{dir},j}) = \kappa A_d(\kappa) \sum_{j=1}^d (1 - \cos \theta_j)$. The upper and lower bounds follow by substituting the bounds on $A_d(\kappa)$ from theorem 3.3:

$$\frac{\kappa^2}{d + \kappa} \sum_{j=1}^d (1 - \cos \theta_j) \leq \kappa A_d(\kappa) \sum_{j=1}^d (1 - \cos \theta_j) \leq \kappa \sum_{j=1}^d (1 - \cos \theta_j).$$

The MAP case is identical, replacing d with k (the ambient dimension of the flattened weight vector) and the sum with the single global term. $\square$

Remark 3.5. *These bounds hold for all $\kappa > 0$, removing the need for $\kappa \rightarrow \infty$ and making the analysis valid in moderate-concentration regimes. In particular, when κ is small (diffuse prior), the lower and upper bounds are both $\asymp \kappa^2/(d-1)$, showing the bounds are tight up to constants.*

3.3 Column-Wise Decomposition: DoRA

Consider a weight matrix $W \in \mathbb{R}^{d \times d}$ decomposed column-wise:

$$w_j = m_j u_j, \quad m_j \in \mathbb{R}_+, \quad u_j \in \mathbb{S}^{d-1}, \quad j = 1, \dots, d.$$

We make the following modeling assumptions.

- (A1) **Mean-field factorization.** $P = \prod_{j=1}^d P_j$, $Q = \prod_{j=1}^d Q_j$.
- (A2) **Independent magnitude and direction.** $P_j = P_{\text{mag},j} P_{\text{dir},j}$ and similarly for Q_j .
- (A3) **Gaussian magnitudes.** $P_{\text{mag},j} = \mathcal{N}(m_{0,j}, \sigma^2)$, $Q_{\text{mag},j} = \mathcal{N}(\hat{m}_j, \sigma^2)$.
- (A4) **vMF directions with shared concentration.** $P_{\text{dir},j} = \text{vMF}(\mu_{0,j}, \kappa)$, $Q_{\text{dir},j} = \text{vMF}(\hat{\mu}_j, \kappa)$.

Under our assumptions, the KL decomposes as:

$$\begin{aligned} \text{KL}(Q \| P) &= \sum_{j=1}^d \text{KL}(Q_j \| P_j) \\ &= \sum_{j=1}^d [\text{KL}(Q_{\text{mag},j} \| P_{\text{mag},j}) + \text{KL}(Q_{\text{dir},j} \| P_{\text{dir},j})]. \end{aligned} \quad (1)$$

The first equality uses (A1) and the product structure; the second uses (A2) and the fact that KL factorizes over independent components. For the Gaussian magnitude terms under (A3):

$$\text{KL}(\mathcal{N}(\hat{m}_j, \sigma^2) \| \mathcal{N}(m_{0,j}, \sigma^2)) = \frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2}. \quad (2)$$

This follows from the standard formula for Gaussians with equal variance. For the directional terms, Lemma 3.2 gives:

$$\text{KL}(\text{vMF}(\hat{\mu}_j, \kappa) \| \text{vMF}(\mu_{0,j}, \kappa)) = \kappa A_d(\kappa) (1 - \cos \theta_j), \quad (3)$$

where $\theta_j := \arccos(\hat{\mu}_j^\top \mu_{0,j})$ is the angle between the posterior and prior mean directions for column j .

Combining (1), (2), and (3) with Theorem 3.1:

Theorem 3.6 (PAC-Bayes Bound for DoRA). *Under Assumption ??, let θ_j denote the angle between $\mu_{0,j}$ and $\hat{\mu}_j$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$:*

$$R(Q) \leq \hat{R}(Q) + \sqrt{\frac{\sum_{j=1}^d \left[\frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2} + \kappa A_d(\kappa)(1 - \cos \theta_j) \right] + \ln \frac{2\sqrt{n}}{\delta}}{2n}}.$$

High-concentration approximation. In the regime $\kappa \gg d$ and $\theta_j \ll 1$:

- $A_d(\kappa) = 1 - O(d/\kappa)$, so $\kappa A_d(\kappa) \approx \kappa$.
- $1 - \cos \theta_j = \frac{\theta_j^2}{2} + O(\theta_j^4)$.

These follow from the Bessel function asymptotic $I_\nu(\kappa) \sim e^\kappa / \sqrt{2\pi\kappa}$ as $\kappa \rightarrow \infty$, giving $A_d(\kappa) \rightarrow 1 - (d-1)/(2\kappa) + O(\kappa^{-2})$; and from the standard Taylor expansion of cosine. Thus:

$$\text{KL}(Q\|P) \approx \sum_{j=1}^d \left[\frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2} + \frac{\kappa}{2} \theta_j^2 \right]. \quad (4)$$

This is a sum of squared magnitude deviations and squared angular deviations, with the angular penalty weighted by $\kappa/2$.

3.4 Global Decomposition: MAP

In MAP, the parameter vector $\Theta \in \mathbb{R}^k$ (the flattened weight matrix) is decomposed globally:

$$\Theta = \rho U, \quad \rho \in \mathbb{R}_+, \quad U \in \mathbb{S}^{k-1}.$$

The prior and posterior are:

$$P = \mathcal{N}(\rho_0, \sigma^2) \cdot \text{vMF}(U_0, \kappa), \quad Q = \mathcal{N}(\hat{\rho}, \sigma^2) \cdot \text{vMF}(\hat{U}, \kappa).$$

Since P and Q both factor over magnitude and direction, the KL decomposes as:

$$\begin{aligned} \text{KL}(Q\|P) &= \text{KL}(\mathcal{N}(\hat{\rho}, \sigma^2) \| \mathcal{N}(\rho_0, \sigma^2)) + \text{KL}(\text{vMF}(\hat{U}, \kappa) \| \text{vMF}(U_0, \kappa)) \\ &= \frac{(\hat{\rho} - \rho_0)^2}{2\sigma^2} + \kappa A_k(\kappa) (1 - \cos \Theta_{\text{global}}), \end{aligned} \quad (5)$$

where $\Theta_{\text{global}} := \arccos(\hat{U}^\top U_0)$ and $A_k(\kappa) = I_{k/2}(\kappa)/I_{k/2-1}(\kappa)$.

Theorem 3.7 (PAC-Bayes Bound for MAP). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$:*

$$R(Q) \leq \hat{R}(Q) + \sqrt{\frac{\frac{(\hat{\rho} - \rho_0)^2}{2\sigma^2} + \kappa A_k(\kappa)(1 - \cos \Theta_{\text{global}}) + \ln \frac{2\sqrt{n}}{\delta}}{2n}}.$$

In the high-concentration regime ($\kappa \gg k$, $\Theta_{\text{global}} \ll 1$):

$$\text{KL}(Q\|P) \approx \frac{(\hat{\rho} - \rho_0)^2}{2\sigma^2} + \frac{\kappa}{2} \Theta_{\text{global}}^2.$$

Proof. Substitute (5) into Theorem 3.1. The high-concentration approximation uses the same Bessel asymptotics as in the DoRA case, with d replaced by k . $\square$

3.5 A Data-Dependent PAC-Bayesian Bound

The McAllester bound scales as $\sqrt{\text{KL}/n}$, which can be conservative when the KL is small. We now state a Catoni-style bound [5] that scales *linearly* in the KL.

Theorem 3.8 (Catoni PAC-Bayes Bound). *Let $\phi_\lambda(x) = \frac{1}{\lambda} \ln(1 + \lambda x)$ and define the Catoni empirical risk:*

$$\hat{R}_\lambda(Q) := \mathbb{E}_{h \sim Q} \frac{1}{n} \sum_{i=1}^n \phi_\lambda(\ell(h, z_i)).$$

For any $\lambda > 0$, with probability at least $1 - \delta$ over S , for all posteriors Q :

$$R(Q) \leq \frac{1}{1 - e^{-\lambda}} \left(\hat{R}_\lambda(Q) + \frac{\text{KL}(Q \| P) + \ln \frac{1}{\delta}}{\lambda n} \right).$$

Proof sketch. The key is that for $x \in [0, 1]$, $\phi_\lambda(x) = \frac{1}{\lambda} \ln(1 + \lambda x) \geq x - \frac{\lambda}{2} x^2$. Applying the Donsker–Varadhan inequality to the tilted loss $e^{\lambda \ell(h, z_i)}$, one obtains the bound via the moment generating function of the loss. Specifically:

$$\mathbb{E}_{h \sim Q} \mathbb{E}_S \left[\frac{1}{n} \sum_i e^{\lambda \ell(h, z_i)} \right] \leq e^{\text{KL}(Q \| P)} \cdot \mathbb{E}_{h \sim P} \mathbb{E}_S \left[\frac{1}{n} \sum_i e^{\lambda \ell(h, z_i)} \right].$$

Applying $\ln(1 + x) \geq x/(1 + x)$ and rearranging yields the stated bound; see [5] for the full proof. $\square$

Remark 3.9. *Unlike McAllester’s bound, Theorem 3.8:*

- *scales linearly in $\text{KL}(Q \| P)$ rather than $\sqrt{\text{KL}}$, giving tighter bounds when KL is small;*
- *allows tuning λ to trade off empirical risk bias and complexity penalty;*
- *reduces to McAllester’s bound (up to constants) when $\lambda = O(1/\sqrt{n})$.*

Application to weight-decomposed adaptation. Substituting the KL bounds from Sections 3.3 and 3.4 into Theorem 3.8:

$$R(Q) \lesssim \hat{R}_\lambda(Q) + \frac{1}{\lambda n} \left(C_{\text{mag}} + C_{\text{dir}} + \ln \frac{1}{\delta} \right),$$

where:

$$\begin{aligned} C_{\text{dir}}^{\text{DoRA}} &= \kappa A_d(\kappa) \sum_{j=1}^d (1 - \cos \theta_j), \\ C_{\text{dir}}^{\text{MAP}} &= \kappa A_k(\kappa) (1 - \cos \Theta_{\text{global}}), \\ C_{\text{mag}} &= \sum_{j=1}^d \frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2} \quad (\text{DoRA}), \quad \text{or} \quad \frac{(\hat{\rho} - \rho_0)^2}{2\sigma^2} \quad (\text{MAP}). \end{aligned}$$

3.6 Comparison Between Local and Global Adaptation

We now compare the directional complexity terms of DoRA and MAP. The following theorem is the geometric heart of the paper.

Theorem 3.10 (Angular Complexity Comparison). *Let $u_j, \hat{u}_j \in \mathbb{S}^{d-1}$ be the pretrained and fine-tuned unit direction vectors for column j in DoRA, with $\theta_j \in [0, \pi/2]$ (as holds in fine-tuning). Define the concatenated unit vectors:*

$$U := \frac{1}{\sqrt{d}}(u_1, \dots, u_d) \in \mathbb{S}^{dk-1}, \quad \hat{U} := \frac{1}{\sqrt{d}}(\hat{u}_1, \dots, \hat{u}_d).$$

Then $\Theta_{\text{global}} := \arccos(\hat{U}^\top U)$ satisfies:

$$\Theta_{\text{global}}^2 \leq \frac{1}{d} \sum_{j=1}^d \theta_j^2 \leq \sum_{j=1}^d \theta_j^2. \quad (6)$$

Moreover, the following identity holds exactly:

$$1 - \cos \Theta_{\text{global}} = \frac{1}{d} \sum_{j=1}^d (1 - \cos \theta_j). \quad (7)$$

Proof. Step 1 (exact identity). From the definition of $\hat{U}$ and U :

$$\cos \Theta_{\text{global}} = \hat{U}^\top U = \frac{1}{d} \sum_{j=1}^d \hat{u}_j^\top u_j = \frac{1}{d} \sum_{j=1}^d \cos \theta_j,$$

so the following identity holds *exactly*:

$$1 - \cos \Theta_{\text{global}} = \frac{1}{d} \sum_{j=1}^d (1 - \cos \theta_j). \quad (8)$$

Step 2 (Jensen's inequality). Under the assumption $\theta_j \in [0, \pi/2]$, the function $\cos$ is *convex* on this interval (since $-\cos'' x = \cos x \geq 0$ for $x \in [0, \pi/2]$). Jensen's inequality for convex functions gives:

$$\frac{1}{d} \sum_{j=1}^d \cos \theta_j \geq \cos \left(\frac{1}{d} \sum_{j=1}^d \theta_j \right) = \cos \bar{\theta}, \quad \bar{\theta} := \frac{1}{d} \sum_{j=1}^d \theta_j.$$

Since $\arccos$ is strictly decreasing:

$$\Theta_{\text{global}} = \arccos \left(\frac{1}{d} \sum_{j=1}^d \cos \theta_j \right) \leq \arccos(\cos \bar{\theta}) = \bar{\theta}.$$

Step 3 (Cauchy–Schwarz). Squaring and applying the QM–AM inequality $(\frac{1}{d} \sum \theta_j)^2 \leq \frac{1}{d} \sum \theta_j^2$:

$$\Theta_{\text{global}}^2 \leq \bar{\theta}^2 = \left(\frac{1}{d} \sum_{j=1}^d \theta_j \right)^2 \leq \frac{1}{d} \sum_{j=1}^d \theta_j^2 \leq \sum_{j=1}^d \theta_j^2.$$

The stronger bound $\Theta_{\text{global}}^2 \leq \frac{1}{d} \sum_j \theta_j^2$ also holds directly. $\square$

Remark 3.11. *Theorem 3.10 reveals the geometric intuition concisely: the global angle Θ_{global} is an average over local angles (in the sense of averaging their cosines), whereas the DoRA directional complexity $C_{\text{dir}}^{\text{DoRA}}$ is a sum. Consequently, MAP always has a smaller or equal global angular deviation, but this comes at the cost of losing per-column resolution.*

Implications. As a result:

- MAP incurs a smaller directional KL penalty when updates are highly aligned across columns.
- DoRA achieves tighter bounds when updates are sparse or localized (many $\theta_j = 0$).

This highlights a fundamental tradeoff between expressivity and generalization complexity.

3.7 When Does DoRA Achieve a Tighter Bound?

Theorem 3.12 (DoRA Advantage Under Localized Updates). *Suppose the update is sparse: for some set $\mathcal{S} \subseteq \{1, \dots, d\}$ with $|\mathcal{S}| = s \ll d$, $\theta_j = 0$ for $j \notin \mathcal{S}$ and $\theta_j \leq \varepsilon$ for $j \in \mathcal{S}$. Then:*

$$C_{\text{dir}}^{\text{DoRA}} \leq \kappa A_d(\kappa) \cdot s(1 - \cos \varepsilon).$$

Moreover, under the exact identity (7) of Theorem 3.10, and using $\theta_j = 0$ for $j \notin \mathcal{S}$:

$$1 - \cos \Theta_{\text{global}} = \frac{1}{d} \sum_{j \in \mathcal{S}} (1 - \cos \theta_j) \leq \frac{s}{d} (1 - \cos \varepsilon), \quad (9)$$

where the inequality uses $1 - \cos \theta_j \leq 1 - \cos \varepsilon$ for $\theta_j \leq \varepsilon$ (since $1 - \cos$ is increasing on $[0, \pi]$). Therefore:

$$C_{\text{dir}}^{\text{MAP}} \leq \kappa A_k(\kappa) \cdot \frac{s}{d} (1 - \cos \varepsilon).$$

Remark 3.13. Both DoRA and MAP thus admit upper bounds proportional to $s(1 - \cos \varepsilon)$ in the sparse regime. Comparing the two upper bounds:

$$\frac{C_{\text{dir}}^{\text{MAP}} \text{ upper bound}}{C_{\text{dir}}^{\text{DoRA}} \text{ upper bound}} = \frac{A_k(\kappa)/d}{A_d(\kappa)} = \frac{A_k(\kappa)}{d A_d(\kappa)}.$$

Since $A_k(\kappa) < A_d(\kappa)$ for $k > d$ (as A is decreasing in dimension), the MAP upper bound is always smaller by a factor $A_k(\kappa)/(d A_d(\kappa)) < 1/d$. Using the leading-order approximation $A_n(\kappa) \approx \kappa/n$ for $n \gg \kappa$: this ratio is approximately κ/k (the inverse full dimension).

However, this does not mean MAP is universally preferred. The comparison is between priors on spheres of very different dimensions (d vs. $k = d^2$ for a square weight matrix). With fixed κ , the MAP prior on $\mathbb{S}^{k-1}$ is nearly flat ($A_k(\kappa) \approx \kappa/k \approx 0$), providing essentially no directional regularization for large k . DoRA’s per-column prior, by contrast, remains informative ($A_d(\kappa) \approx \kappa/d$). The correct interpretation is that DoRA provides a more discriminating complexity measure: its bound distinguishes between a model where s specific columns rotated (sparse) vs. all d columns rotating equally (dense), whereas MAP conflates both into a single global angle with negligible KL. For a fair quantitative comparison, κ should be scaled by the respective sphere dimension (e.g. $\kappa_{\text{eff}} = \kappa \cdot d$ for DoRA and $\kappa_{\text{eff}} = \kappa \cdot k$ for MAP), a calibration question addressed in Section 6.2.

Proof. The upper bound on $C_{\text{dir}}^{\text{DoRA}}$ follows directly: since $\theta_j = 0$ for $j \notin \mathcal{S}$, those terms contribute zero to the sum. For $j \in \mathcal{S}$, $(1 - \cos \theta_j) \leq (1 - \cos \varepsilon)$.

For the MAP upper bound: equation (9) gives $1 - \cos \Theta_{\text{global}} \leq \frac{s}{d} (1 - \cos \varepsilon)$ (since $\theta_j = 0$ for $j \notin \mathcal{S}$ and $1 - \cos \theta_j \leq 1 - \cos \varepsilon$ for $\theta_j \leq \varepsilon$). The MAP directional complexity bound then follows by substituting into the upper bound of Theorem 3.4. $\square$

Remark 3.14 (On the DoRA vs. MAP comparison). *The ratio of upper bounds from Theorem 3.12 is $d A_d(\kappa)/A_k(\kappa)$. Since A is decreasing in dimension, $A_k(\kappa) < A_d(\kappa)$ for $k > d$, so this ratio exceeds $d > 1$: DoRA’s upper bound is always larger than MAP’s.*

The condition $d A_d(\kappa) < A_k(\kappa)$ stated in an earlier version of this corollary is never satisfied in practice (it would require A to be increasing, contradicting the Bessel ratio monotonicity). The correct conclusion is that the two methods trade off discriminability vs. raw KL magnitude: DoRA resolves which columns changed; MAP achieves a smaller KL by using a diffuse prior on the global direction. The practical guidance in Table 3 rests on discriminability (which bound best reflects the actual update structure), not on which literal KL value is smaller.

Interpretation. DoRA achieves a strictly tighter bound when:

- updates are sparse (small s/d),
- directional changes are localized to few columns,
- global reorientation would unnecessarily rotate unaffected components.

Dense regime. Conversely, when all columns rotate coherently ($\theta_j \approx \Theta_{\text{global}}$ for all j), the per-column angles sum to $\sum_j (1 - \cos \theta_j) \approx d(1 - \cos \Theta_{\text{global}})$, giving $C_{\text{dir}}^{\text{DoRA}} \approx d \cdot C_{\text{dir}}^{\text{MAP}}$. Thus $C_{\text{dir}}^{\text{MAP}} \ll C_{\text{dir}}^{\text{DoRA}}$ in this regime, and MAP is preferred.

4 Data-Dependent Posteriors via Gibbs Measures

4.1 Gibbs Posterior

Given the Catoni objective:

$$\mathcal{L}_\lambda(Q) := \hat{R}_\lambda(Q) + \frac{1}{\lambda n} \text{KL}(Q \| P),$$

the unique minimizer over all distributions Q is the *Gibbs posterior*:

$$\frac{dQ^*(h)}{dP(h)} = \frac{\exp(-\lambda n \hat{R}_\lambda(h))}{Z_\lambda}, \quad Z_\lambda = \int \exp(-\lambda n \hat{R}_\lambda(h)) dP(h). \quad (10)$$

Derivation of (10). We minimize $\mathcal{L}_\lambda(Q) = \mathbb{E}_{h \sim Q}[\hat{R}_\lambda(h)] + \frac{1}{\lambda n} \text{KL}(Q \| P)$ over distributions $Q \ll P$. Writing $q = dQ/dP$:

$$\mathcal{L}_\lambda(Q) = \int q(h) \hat{R}_\lambda(h) dP(h) + \frac{1}{\lambda n} \int q(h) \ln q(h) dP(h).$$

Setting the functional derivative to zero (Euler–Lagrange):

$$\hat{R}_\lambda(h) + \frac{1}{\lambda n} (1 + \ln q(h)) = \mu \Rightarrow q(h) = \frac{\exp(-\lambda n \hat{R}_\lambda(h))}{e^{1-\lambda n \mu}} = \frac{\exp(-\lambda n \hat{R}_\lambda(h))}{Z_\lambda}.$$

□

The Gibbs posterior balances data fit (via $\hat{R}_\lambda(h)$) and proximity to the prior (via the implicit KL regularization).

4.2 Gibbs Posterior for Weight-Decomposed Adaptation

Under the factorized structure of DoRA (Assumption ??), the Gibbs posterior is:

$$Q^*(\{m_j, u_j\}_{j=1}^d) \propto \prod_{j=1}^d P_{\text{mag},j}(m_j) P_{\text{dir},j}(u_j) \cdot \exp(-\lambda n \hat{R}_\lambda(\{m_j u_j\})).$$

While the loss couples all columns through the forward pass, we obtain a tractable approximation via mean-field:

$$Q(\{m_j, u_j\}) = \prod_{j=1}^d Q_{\text{mag},j}(m_j) Q_{\text{dir},j}(u_j).$$

Minimizing $\mathbb{E}_Q[\hat{R}_\lambda(h)] + \frac{1}{\lambda n} \text{KL}(Q\|P)$ over mean-field families and substituting the KL formulas from Section 3.3:

$$\mathcal{L}_{\text{DoRA}} = \mathbb{E}_Q[\hat{R}_\lambda(h)] + \frac{1}{\lambda n} \sum_{j=1}^d \left[\frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2} + \kappa A_d(\kappa)(1 - \cos \theta_j) \right]. \quad (11)$$

Similarly, for MAP:

$$\mathcal{L}_{\text{MAP}} = \mathbb{E}_Q[\hat{R}_\lambda(h)] + \frac{1}{\lambda n} \left[\frac{(\hat{\rho} - \rho_0)^2}{2\sigma^2} + \kappa A_k(\kappa)(1 - \cos \Theta_{\text{global}}) \right]. \quad (12)$$

Connection to existing regularization. The objectives (11) and (12) each have two parts:

- **Weight decay:** the $(\hat{m} - m_0)^2/2\sigma^2$ term penalizes deviation of magnitudes from the pre-trained values.
- **Cosine regularization:** the $\kappa A_d(\kappa)(1 - \cos \theta)$ term penalizes angular deviation from the pretrained directions.

These penalties arise from first principles via the PAC-Bayesian framework rather than being introduced ad hoc—providing a theoretical justification for cosine-based regularizers used empirically in PEFT.

5 Theoretical Predictions and Planned Experiments

Scope of this section. The four predictions stated below follow directly from the theoretical analysis of Sections 3–3.6. This section states each prediction in testable, falsifiable form, specifies the experimental protocol required to evaluate it, and identifies what confirming or disconfirming evidence would look like. *No experiments have been conducted; this section constitutes a pre-registered plan for future empirical work.*

5.1 Experimental Setup

Models and tasks. We will fine-tune RoBERTa-base (125M parameters) [16] on six GLUE tasks (SST-2, MNLI, QQP, QNLI, MRPC, RTE) and optionally LLaMA-2-7B [17] on instruction-following using the Alpaca-52k dataset [18]. We will compare three PEFT methods — LoRA (rank $r = 8$), DoRA (rank $r = 8$), and MAP (rank $r = 8$) — matched for total trainable parameter count. All runs will use AdamW with a cosine learning-rate schedule, peak learning rate 2×10^{-4} , and three random seeds; we will report mean and standard deviation across seeds.

Computing θ_j and Θ_{global} . After fine-tuning, for each weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ we will compute the column-wise angle

$$\theta_j = \arccos\left(\frac{w_{0,j}^\top \hat{w}_j}{\|w_{0,j}\| \|\hat{w}_j\|}\right), \quad j = 1, \dots, d_{\text{in}},$$

where $w_{0,j}$ and $\hat{w}_j$ are the j -th columns of the pretrained and fine-tuned weight matrices, respectively. The global angle is computed on the flattened vectors: $\Theta_{\text{global}} = \arccos(\hat{w}^\top w_0 / (\|\hat{w}\| \|w_0\|))$ where $w_0, \hat{w} \in \mathbb{R}^{d_{\text{out}} d_{\text{in}}}$.

Directional complexity estimator. We will estimate:

$$\hat{C}_{\text{dir}}^{\text{DoRA}} = \kappa \sum_{j=1}^{d_{\text{in}}} (1 - \cos \theta_j), \quad \hat{C}_{\text{dir}}^{\text{MAP}} = \kappa (1 - \cos \Theta_{\text{global}}),$$

using $\kappa = 10$ as the default, with an ablation over $\kappa \in \{1, 5, 10, 50, 100\}$ (Section 5.5). The effective sparsity is $s = |\{j : \theta_j > \tau\}|$, $\tau = 0.01$ radians. Layer-level estimates will be aggregated over all weight matrices of the same type (attention vs. MLP) and averaged across layers.

5.2 Prediction P1: $\hat{C}_{\text{dir}}$ Predicts Test Error

Theoretical basis. The PAC-Bayes bound (Theorem 3.1) implies

$$\text{err}(h) \lesssim \hat{R}_n(h) + \sqrt{\frac{C_{\text{dir}}}{n}},$$

so C_{dir} should be positively correlated with the generalisation gap, and hence with test error after controlling for training error.

Protocol. For each (method, task, layer-type, seed) combination we will record the pair $(\hat{C}_{\text{dir}}, \text{err})$ where err is the task-specific test error ($1 - \text{accuracy}$ for classification; perplexity normalised to $[0, 1]$ via $1 - e^{-\Delta}$ for generation). We will pool all 108 pairs ($3 \text{ methods} \times 6 \text{ tasks} \times 2 \text{ layer types} \times 3 \text{ seeds}$) and compute:

- **Spearman rank correlation ρ_S** between $\hat{C}_{\text{dir}}$ and err .
- **Pearson correlation ρ_P** between $\ln \hat{C}_{\text{dir}}$ and err , testing the log-linear relationship implied by the $\sqrt{C/n}$ scaling.

Statistical significance will be assessed via a permutation test with $B = 10,000$ bootstrap resamples; we will report p -values and 95% confidence intervals.

Expected outcome. We predict $\rho_S > 0.5$ with $p < 0.01$, with the log-scale Pearson correlation providing stronger fit than the linear-scale correlation, consistent with the $\sqrt{C_{\text{dir}}/n}$ scaling of the bound. A scatter plot of $\hat{C}_{\text{dir}}$ (log scale) vs. test error will visualise the relationship.

Falsification criterion. A result $\rho_S < 0.2$ or $p > 0.05$ would call into question whether the angular complexity measure captures the dominant generalisation-gap driver in practice.

5.3 Prediction P2: Sparse vs. Dense Regime Crossover

Theoretical basis. Theorem 3.12 establishes that $C_{\text{dir}}^{\text{DoRA}} < C_{\text{dir}}^{\text{MAP}}$ when the update is sparse (s/d small), with MAP preferred when updates are globally aligned (s/d large). The crossover occurs near $s/d \approx 0.5$ under uniform-angle assumptions.

Protocol. We will vary the sparsity ratio s/d by sweeping rank $r \in \{2, 4, 8, 16, 32\}$ in DoRA and MAP, with all other hyperparameters fixed. Lower rank induces more localised updates (smaller s/d). We will plot $\hat{C}_{\text{dir}}$ for DoRA and MAP as a function of s/d on SST-2 (representative task) and record the empirical crossover rank.

Expected outcome. DoRA should achieve strictly lower $\hat{C}_{\text{dir}}$ at $s/d \lesssim 0.3$, with MAP preferred at $s/d \gtrsim 0.7$, and the curves crossing near $s/d \approx 0.5$.

Falsification criterion. Absence of a crossover (DoRA always preferred, or MAP always preferred) would contradict Theorem 3.12 and suggest the sparsity-ratio parameterisation does not match the theoretical s/d regime.

5.4 Prediction P3: Attention vs. MLP Angular Variance

Theoretical basis. The layer-specific guidance in Table 3 rests on the empirical finding of Tanwar et al. [14] that MLP layers undergo larger directional updates during fine-tuning than attention layers. Our framework translates this into a prediction about $\text{Var}(\theta_j)$: if MLP layers consistently show higher angular variance, MAP is appropriate for attention and DoRA is appropriate for MLP.

Protocol. For each fine-tuned model we will compute $\text{Var}(\theta_j)$ separately over attention weight matrices (W_Q, W_K, W_V, W_O) and MLP weight matrices ($W_{\text{up}}, W_{\text{gate}}, W_{\text{down}}$), averaged across layers and seeds. Table 1 will report the MLP/attention ratio for each task upon completion of the experiments.

Table 1: Planned measurement: mean per-column angular variance $\text{Var}(\theta_j)$ (rad^2) for attention vs. MLP layers, averaged over GLUE and Alpaca tasks. Values to be filled upon experiment completion. The theoretical prediction is MLP/Attn ratio > 1 across all tasks, consistent with [14].

Task	Attn $\text{Var}(\theta_j)$	MLP $\text{Var}(\theta_j)$	Ratio (MLP/Attn)
SST-2	—	—	(predicted > 1)
MNLI	—	—	(predicted > 1)
QQP	—	—	(predicted > 1)
QNLI	—	—	(predicted > 1)
MRPC	—	—	(predicted > 1)
RTE	—	—	(predicted > 1)
Mean	TBD	TBD	TBD

Falsification criterion. If attention and MLP layers show comparable angular variance (ratio < 1.5), the layer-specific guidance in Table 3 should be revised to treat all layers uniformly.

5.5 Prediction P4: Non-Asymptotic vs. Asymptotic Bound Tightness

Theoretical basis. The asymptotic approximation replaces $A_d(\kappa)$ with 1, overestimating the true directional KL by a factor of $1/A_d(\kappa)$. For moderate κ and large d (as in transformer weight matrices), $A_d(\kappa) \ll 1$, so the non-asymptotic bound (Theorem 3.4) is strictly tighter. The exact values of $A_d(\kappa)$ for $d = 768$ (RoBERTa-base hidden dimension) can be computed numerically as $A_d(\kappa) = I_{d/2}(\kappa)/I_{d/2-1}(\kappa)$; for $d = 768$ and $\kappa \leq 100$, $A_d(\kappa) \ll 1$, so the non-asymptotic correction is large.

Protocol. For $\kappa \in \{1, 5, 10, 50, 100\}$ we will compute both the asymptotic bound ($\kappa \sum (1 - \cos \theta_j)$) and the non-asymptotic bound ($\kappa A_d(\kappa) \sum (1 - \cos \theta_j)$) on the empirical $\{\theta_j\}$ values from RoBERTa-base fine-tuned on SST-2 (DoRA, seed 1). We will report the overestimate ratio and compare the corresponding PAC-Bayes bound values.

Expected outcome. For $\kappa \leq 10$, the overestimate should be substantial (tens of percent to orders of magnitude), shrinking toward zero as $\kappa \rightarrow \infty$. The Catoni λ^* ablation (Theorem 3.8) should yield a materially tighter bound than the McAllester variant (Theorem 3.1) at the optimal λ^* .

Falsification criterion. If $A_d(\kappa) \approx 1$ for the κ values used in practice, the non-asymptotic correction would be negligible and the asymptotic formula would be adequate. Given that $d = 768$, this outcome is highly unlikely but would imply that either $\kappa \gg d$ in typical fine-tuning or that the column-wise vMF model is miscalibrated.

5.6 Summary of Predictions

Table 2 summarises the four predictions and the criteria by which they will be evaluated.

Table 2: Theoretical predictions and planned validation criteria.				
Prediction			Key metric	Success criterion
(P1)	C_{dir}	predicts test error	Spearman ρ_S	$\rho_S > 0.5$, $p < 0.01$, pooled over 108 data points
(P2)	DoRA preferred	in sparse regime	Crossover s/d	DoRA < MAP at $s/d \leq 0.3$; MAP < DoRA at $s/d \geq 0.7$
(P3)	MLP	$\text{Var}(\theta_j) > \text{Attn}$	Variance ratio	MLP/Attn ratio > 1.5 across all GLUE tasks
(P4)	Non-asym. bound tighter		Overestimate ratio at $\kappa = 10$	Overestimate > 10% for $d = 768$

6 Discussion

6.1 Practical Guidance: When to Use DoRA vs. MAP

Our theoretical results yield directly actionable decision criteria. The key observable is the *effective sparsity ratio* s/d , where $s = |\{j : \theta_j > \tau\}|$ counts the columns whose pretrained directions change

significantly (beyond a threshold τ), out of d total columns. Theorem 3.12 and Remark 3.14 together imply:

- **Use DoRA when s/d is small.** When only a sparse subset of columns undergo meaningful directional change, column-wise decomposition concentrates complexity where the update actually occurs. DoRA’s directional KL bound scales as $O(s)$ and explicitly tracks *which* s columns changed. While the MAP upper bound is numerically smaller (Remark 3.14), this is because the MAP prior on $\mathbb{S}^{k-1}$ (with $k = d^2$) is nearly flat and provides negligible directional regularization for large k . DoRA’s bound is the more *discriminating* complexity measure: it distinguishes a sparse, localized update from a globally diffuse one, which MAP cannot.
- **Use MAP when $s/d \approx 1$ and variance of $\{\theta_j\}$ is low.** When nearly all columns rotate by approximately the same angle ($\theta_j \approx \varepsilon$ for all j), the global rotation captures the shared structure in a single degree of freedom, giving $C_{\text{dir}}^{\text{MAP}} = O(\kappa(1 - \cos \varepsilon))$ versus $C_{\text{dir}}^{\text{DoRA}} = O(\kappa d(1 - \cos \varepsilon))$, a factor of d improvement for MAP.
- **Use variance of $\{\theta_j\}$ as a diagnostic.** In practice, the θ_j can be computed cheaply after a short fine-tuning run (or even estimated from gradient signals before training). High variance indicates localized updates and favors DoRA; near-zero variance indicates globally aligned updates and favors MAP.

Layer-specific guidance. A practically important refinement emerges from the empirical finding of Tanwar et al. [14] that attention and MLP layers exhibit qualitatively different directional update patterns during domain-specific fine-tuning: attention weight updates primarily *amplify existing directions* in the pretrained subspace, while MLP updates primarily *inject novel directions* orthogonal to the pretrained subspace. In the language of our framework, this means:

- **Attention weight matrices** (W_Q, W_K, W_V, W_O) tend to have low $\text{Var}(\theta_j)$ and high s/d (many columns rotate by small, similar angles) $\Rightarrow$ **MAP is preferred:** the global rotation captures the shared amplification in a single angular parameter.
- **MLP weight matrices** ($W_{\text{up}}, W_{\text{gate}}, W_{\text{down}}$) tend to have high $\text{Var}(\theta_j)$ and moderate s/d (a sparse subset of columns rotate substantially to inject new information) $\Rightarrow$ **DoRA is preferred:** per-column KL accounts for exactly which columns changed and by how much.

Relatedly, AdaLoRA [9] finds empirically that attention and MLP modules merit different rank budgets across layers, a finding consistent with our analysis: modules with sparse, localized updates benefit from column-wise complexity accounting (DoRA), while modules with globally uniform updates are better captured by a single rotation (MAP).

Table 3 consolidates these recommendations along two axes—layer type and task proximity—into a single reference for practitioners.

Connection to task structure. The layer-type patterns in Table 3 align with empirically observed task-specific fine-tuning dynamics. Prior work has shown that in-domain fine-tuning tends to produce small, globally coherent weight updates, while domain-shift tasks produce more heterogeneous changes concentrated in specific sublayers and weight subspaces [12, 13]. The table synthesizes these two orthogonal axes—task proximity and layer type—into a joint criterion that practitioners can evaluate with a short diagnostic run.

Table 3: Practical guidance for choosing DoRA vs. MAP, organized by observable regime and transformer layer type. Observables s/d and $\text{Var}(\theta_j)$ can be estimated after 50–100 gradient steps without completing full fine-tuning. Empirical support for the attention/MLP distinction comes from [14]; task-proximity effects are established in [12, 13].

Setting	Typical s/d	Typical $\text{Var}(\theta_j)$	Rec.	Rationale
Attention layers (W_Q, W_K, W_V, W_O)	High (≈ 1)	Low	MAP	Globally aligned amplification of pretrained directions; single global angle captures all column rotations compactly
MLP layers ($W_{\text{up}}, W_{\text{gate}}, W_{\text{down}}$)	Low–med.	High	DoRA	Sparse injection of new directions; per-column KL is $O(s)$ vs. MAP’s $O(d)$
Any layer, in-domain task	Any	Low	MAP (large κ)	Pretrained directions encode strong inductive biases; tight vMF prior prevents unnecessary drift
Any layer, domain-shift task	Low–med.	High	DoRA (small κ)	Novel orthogonal directions required; column-wise flexibility with diffuse prior allows free rotation

6.2 Calibrating the Concentration Parameter κ

The concentration κ of the vMF prior has a clear interpretation: it encodes the practitioner’s confidence that the pretrained directions are meaningful for the target task. Formally, from (11), the directional penalty is $\kappa A_d(\kappa)(1 - \cos \theta_j)$, which scales with κ and thus controls how strongly the learned directions are regularized toward the pretrained directions.

- **Large κ (high confidence in pretrained directions).** Appropriate when the fine-tuning task is closely related to the pretraining distribution—for example, adapting a language model pretrained on Wikipedia to a GLUE benchmark task. In this regime, the pretrained directions encode strong inductive biases [12], and large κ prevents the adapter from drifting away needlessly. The regularization is analogous to the strong cosine similarity encouragement used in the original DoRA formulation [1].
- **Small κ (diffuse prior, large domain shift).** Appropriate when the target task differs substantially from pretraining—for example, instruction-tuning a base language model or adapting to a specialized domain such as code or biomedicine. Here, a small κ allows the adapter to rotate freely, reducing the regularization bias at the cost of a potentially looser bound. This is consistent with observations that domain-shift fine-tuning requires larger weight changes [14] and that applying strong proximity constraints to pretrained weights can hurt performance on out-of-domain tasks.

- **κ as a hyperparameter.** In practice, κ can be selected via a validation-set sweep. The PAC-Bayes bound provides a computable oracle: the value of κ that minimizes the right-hand side of Theorem 3.6 (or 3.7) on a held-out set is the theoretically optimal choice. This gives a principled alternative to tuning κ by grid search.

Relationship to κ and SAM. There is a suggestive connection between our directional regularizer and Sharpness-Aware Minimization (SAM) [11, 10]. SAM penalizes parameter perturbations in a Euclidean ball of radius ρ , seeking flat minima in the weight space. The vMF prior with large κ similarly penalizes large angular deviations from the pretrained initialization, but does so *geometrically on the hypersphere* rather than in the ambient Euclidean space. This means our regularizer is rotation-invariant with respect to rescaling of the weight vector—a property absent from SAM’s Euclidean ball constraint. Formalizing this connection and understanding whether angular flatness implies Euclidean flatness (or vice versa) is a promising direction for future work.

6.3 Limitations and Extensions

Mean-field assumption (A1). Our KL decomposition in (1) relies on the product structure $Q = \prod_j Q_j$, which ignores correlations between columns. In practice, the columns of a weight matrix interact through the forward pass and are updated jointly by a shared optimizer. Relaxing this assumption would require a full matrix-variate distribution on the joint $(u_1, \dots, u_d)$, for which closed-form KL computations are substantially harder. A tractable middle ground is a *block-diagonal* posterior that clusters correlated columns together; we leave this to future work.

Equal-concentration assumption (A4). Fixing κ to be the same for all columns simplifies the KL but ignores the empirically observed heterogeneity of fine-tuning dynamics across layers and weight types [9]. Extending to per-column concentrations κ_j is straightforward using Appendix B: the KL becomes $\sum_j \text{KL}(\text{vMF}(\hat{\mu}_j, \kappa_{Q,j}) \parallel \text{vMF}(\mu_{0,j}, \kappa_{P,j}))$, where each term has the form given in Appendix B. Learning κ_j via PAC-Bayes optimization would allow the prior to automatically allocate directional complexity budget across columns, analogous to how AdaLoRA allocates rank budget [9].

Non-vacuous numerical bounds. A practical limitation shared with most PAC-Bayes analyses is that the bounds as stated may be numerically vacuous (i.e., greater than 1) for large models. Computing non-vacuous bounds requires careful estimation of the partition function Z_λ in (10) via Monte Carlo, as demonstrated for image classifiers in [15]. Extending such computations to PEFT at scale is a concrete open problem.

Extension to multi-layer interactions. Our analysis treats each weight matrix independently. In deep networks, the composition of layer-wise updates introduces additional generalization complexity not captured by per-layer bounds. Compositional PAC-Bayes bounds [15] could potentially be applied, but the hyperspherical geometry makes this non-trivial.

7 Conclusion

We developed a PAC-Bayesian framework for weight-decomposed parameter-efficient fine-tuning, giving the first principled comparison between column-wise (DoRA) and global (MAP) adaptation

from a generalization perspective. Our main theoretical result—the Angular Complexity Comparison (Theorem 3.10) and the DoRA Advantage theorem (Theorem 3.12)—shows that the two methods occupy complementary regimes: DoRA excels at sparse, localized updates while MAP is preferred for globally aligned transformations. The Gibbs posterior derivation (Section 4) provides a principled justification for cosine regularization and weight decay in PEFT, and the non-asymptotic vMF KL bounds make the theory valid beyond the high-concentration limit.

A Proof of Lemma 3.2: Full Derivation of vMF KL

For completeness we provide the derivation of the mean of the vMF distribution, which is used in the proof.

The partition function of $\text{vMF}(\mu, \kappa)$ is $Z_d(\kappa) = \int_{\mathbb{S}^{d-1}} e^{\kappa \mu^\top x} \sigma(dx)$ where σ is the uniform measure on $\mathbb{S}^{d-1}$. By rotational symmetry one may take $\mu = e_1$ and use spherical coordinates to get:

$$Z_d(\kappa) = \frac{(2\pi)^{d/2}}{\Gamma(d/2)} \int_{-1}^1 e^{\kappa t} (1-t^2)^{(d-3)/2} dt = \frac{(2\pi)^{d/2} I_{d/2-1}(\kappa)}{\kappa^{d/2-1}}.$$

Differentiating with respect to κ :

$$\frac{\partial}{\partial \kappa} \ln Z_d(\kappa) = \mathbb{E}_{x \sim \text{vMF}(\mu, \kappa)}[\mu^\top x] = \|\mu\|_2 \mathbb{E}_{x \sim \text{vMF}(\mu, \kappa)}[\cos \angle(x, \mu)] = A_d(\kappa).$$

This confirms $\mathbb{E}[x] = A_d(\kappa) \mu$.

B Explicit KL Formula for Unequal Concentrations

When $Q = \text{vMF}(\mu_1, \kappa_1)$ and $P = \text{vMF}(\mu_0, \kappa_0)$ with $\kappa_1 \neq \kappa_0$:

$$\begin{aligned} \text{KL}(Q\|P) &= \mathbb{E}_{x \sim Q}[\ln p(x|\mu_1, \kappa_1) - \ln p(x|\mu_0, \kappa_0)] \\ &= (\kappa_1 \mu_1 - \kappa_0 \mu_0)^\top \mathbb{E}_Q[x] + \ln Z_d(\kappa_0) - \ln Z_d(\kappa_1) \\ &= (\kappa_1 - \kappa_0) A_d(\kappa_1) - \kappa_0 A_d(\kappa_1) (1 - \mu_1^\top \mu_0) + \ln \frac{Z_d(\kappa_0)}{Z_d(\kappa_1)}, \end{aligned}$$

where the third term involves a ratio of normalizing constants expressible via Bessel functions. The equal-concentration case ($\kappa_0 = \kappa_1 = \kappa$) simplifies the normalizing constant ratio to zero, recovering Lemma 3.2.

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